

WAREN, Germany

WMO: 102680

Lat: 53.5197N

Lon: 12.6653E

Elev: 65

StdP: 100.54

Time Zone: 1.00 (EUC)

Period: 94-13

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.4	-8.4	-14.3	1.1	-9.9	-11.3	1.4	-6.1	11.8	5.1	10.2	5.7	2.3	80	0.547

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.3	29.2	18.8	27.6	18.4	25.7	17.8	20.6	26.1	19.6	24.8	18.8	23.4	3.0	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.9	13.8	21.9	18.0	13.0	21.3	17.1	12.3	20.4	59.4	26.3	56.2	24.9	53.6	23.4	31.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.3	8.0	7.0	-13.5	32.7	4.2	1.9	-16.5	34.1	-18.9	35.2	-21.3	36.3	-24.3	37.7		
(5)			-13.8	23.1	4.1	2.1	-16.8	24.7	-19.2	25.9	-21.5	27.1	-24.4	28.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	0.2	1.1	3.7	8.9	13.4	16.2	18.8	18.7	14.6	9.6	4.9	1.0	(6)
(7)		DBStd	7.66	4.65	4.36	3.80	4.08	3.48	3.14	3.11	2.93	3.00	3.52	3.34	4.37	(7)
(8)		HDD10.0	1307	304	248	196	68	8	0	0	0	2	51	153	278	(8)
(9)		HDD18.3	3405	562	481	452	284	157	79	32	32	117	271	402	536	(9)
(10)		CDD10.0	1061	0	0	2	34	114	187	273	271	141	38	2	0	(10)
(11)		CDD18.3	116	0	0	0	1	4	16	46	43	6	0	0	0	(11)
(12)		CDH23.3	985	0	0	0	9	58	149	409	316	42	1	0	0	(12)
(13)		CDH26.7	246	0	0	0	1	8	38	119	76	3	0	0	0	(13)
(14)	Wind	WSAvg	3.3	3.6	3.7	3.6	3.2	3.1	3.1	2.9	2.9	2.9	3.5	3.3	3.8	(14)
(15)	Precipitation	PrecAvg	584	48	38	40	33	50	61	67	59	50	44	45	51	(15)
(16)		PrecMax	732	91	90	94	69	94	119	286	151	150	86	99	106	(16)
(17)		PrecMin	403	0	8	9	2	10	10	10	21	13	21	5	11	(17)
(18)		PrecStd	86	26	17	20	17	24	26	50	28	30	19	21	22	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.0	11.7	16.5	24.7	27.7	30.5	32.1	30.8	26.9	20.0	13.2	11.0	(19)
(20)			MCWB	9.6	9.9	10.4	15.1	17.2	19.4	20.0	19.8	17.8	16.2	11.7	10.1	(20)
(21)		2%	DB	8.9	9.9	13.6	20.6	24.9	27.1	29.7	28.9	24.5	17.6	11.7	9.2	(21)
(22)			MCWB	7.9	8.3	9.2	13.1	16.6	18.4	19.5	18.3	17.1	14.6	10.3	8.2	(22)
(23)		5%	DB	7.4	8.5	11.2	18.3	22.3	24.6	27.7	27.1	22.1	15.9	10.6	7.7	(23)
(24)			MCWB	6.5	7.1	8.1	11.6	15.3	17.2	18.7	18.2	16.2	13.6	9.5	6.9	(24)
(25)		10%	DB	6.1	7.0	9.2	16.0	20.3	22.5	25.5	24.9	19.9	14.4	9.4	6.5	(25)
(26)			MCWB	5.2	5.7	7.0	10.8	14.3	16.1	18.0	17.9	15.3	12.9	8.4	5.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	9.7	10.4	11.5	16.0	19.3	21.3	21.8	21.8	19.2	17.1	12.0	10.5	(27)
(28)			MCDWB	10.7	11.4	14.5	22.0	25.0	27.2	28.0	27.3	24.2	19.9	12.6	10.7	(28)
(29)		2%	WB	8.0	8.6	9.9	14.1	17.6	19.5	20.7	20.3	18.1	15.2	10.6	8.4	(29)
(30)			MCDWB	8.8	9.5	12.4	19.2	23.0	24.8	26.8	25.0	22.2	16.6	11.4	9.1	(30)
(31)		5%	WB	6.6	7.3	8.4	12.6	16.2	18.2	19.7	19.4	17.1	14.1	9.5	7.0	(31)
(32)			MCDWB	7.3	8.3	10.6	16.9	20.7	22.6	25.6	23.7	20.8	15.5	10.3	7.5	(32)
(33)		10%	WB	5.3	5.8	7.2	11.2	15.0	17.0	18.9	18.6	16.1	13.0	8.5	5.8	(33)
(34)			MCDWB	5.9	6.9	8.9	15.0	19.1	21.3	23.8	22.8	19.2	14.4	9.2	6.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	4.2	5.1	6.9	9.2	9.6	9.3	9.3	9.1	8.1	6.4	4.1	3.7	(35)
(36)			MCDBR	4.7	5.9	10.4	13.0	13.5	13.3	13.2	12.7	11.8	7.9	4.8	4.2	(36)
(37)			MCWBR	4.5	4.6	6.5	6.6	6.5	6.0	4.8	4.7	5.6	5.1	3.8	3.8	(37)
(38)		5% WB	MCDBR	4.7	5.8	9.0	11.5	11.6	11.8	11.4	10.6	10.1	7.4	4.5	4.1	(38)
(39)			MCWBR	4.6	4.8	6.2	6.3	6.2	6.2	4.8	4.9	5.2	5.1	3.8	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.314	0.338	0.372	0.402	0.398	0.396	0.415	0.411	0.377	0.360	0.336	0.307	(40)	
(41)		taud	2.318	2.313	2.252	2.237	2.287	2.313	2.279	2.303	2.379	2.392	2.372	2.315	(41)	
(42)		Ebn at Noon	648	753	800	825	850	854	828	808	792	719	618	579	(42)	
(43)		Edh at Noon	60	82	106	123	123	121	123	114	94	75	57	50	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.60	1.26	2.44	4.08	5.09	5.36	5.00	4.21	3.08	1.66	0.72	0.41	(44)	
(45)		RadStd	0.07	0.17	0.28	0.58	0.61	0.42	0.70	0.34	0.33	0.22	0.08	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page